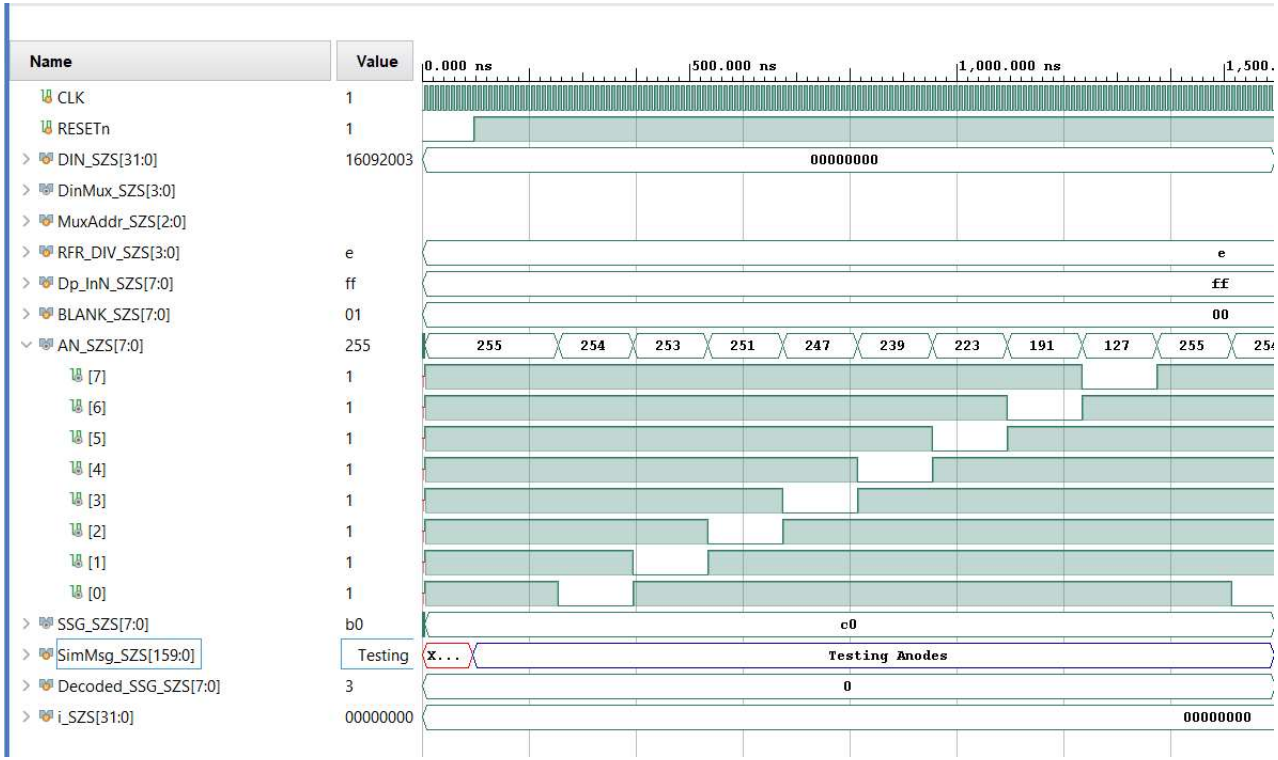
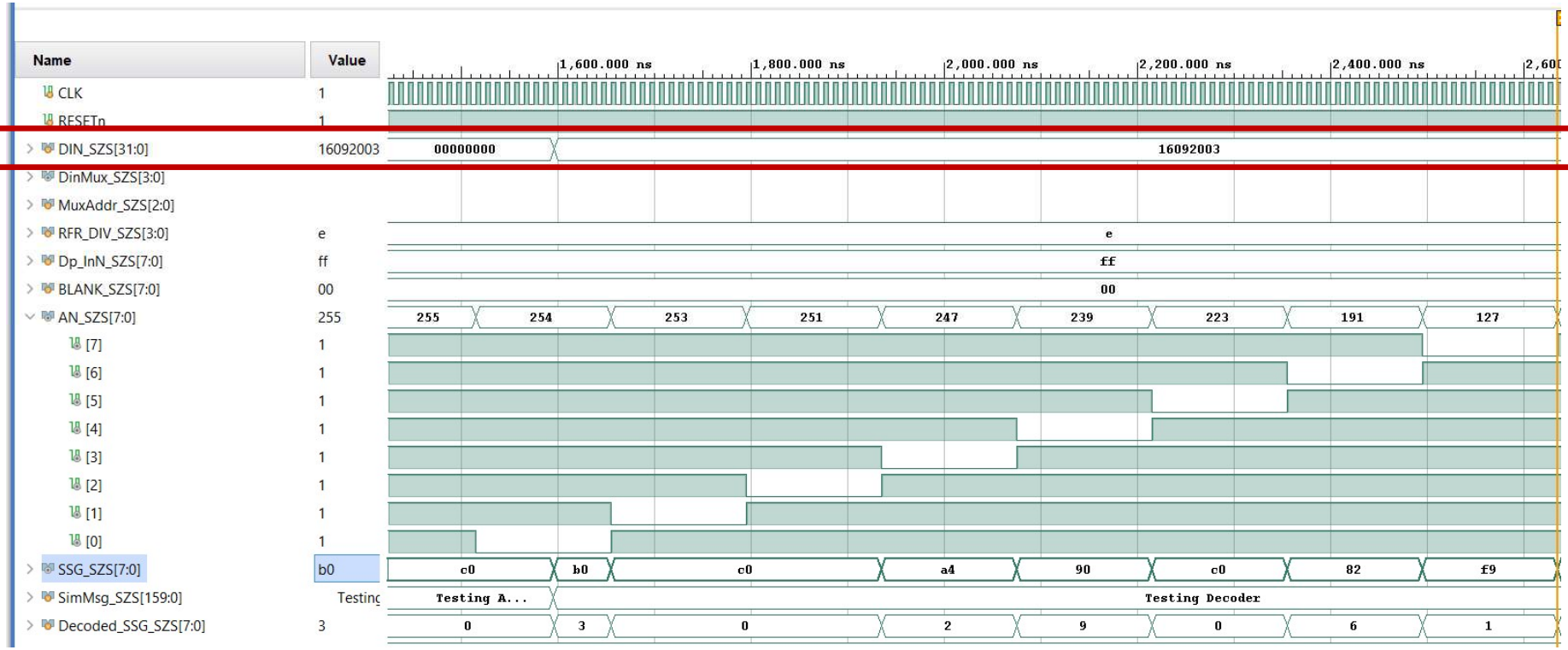
Seven-segment display

Includes anode testing, blanking behavior, seven-segment decoder waveforms, and photographs of the Nexys A7 displaying digits at different refresh rates. This is useful evidence of both simulation and hardware testing.

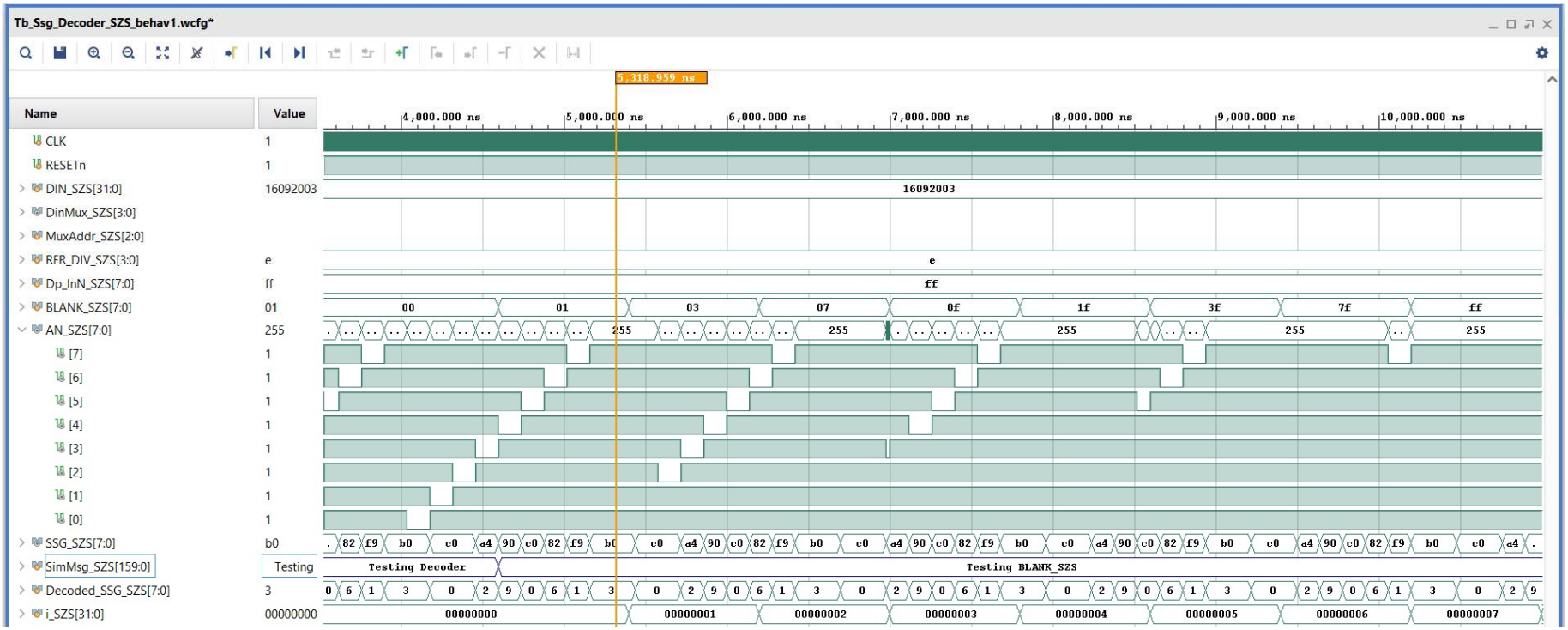
Testing Anodes



Date of birth (16.09.2003)



Blank check simulation tests the blanking functionality of the seven-segment display decoder using a Verilog testbench in Vivado

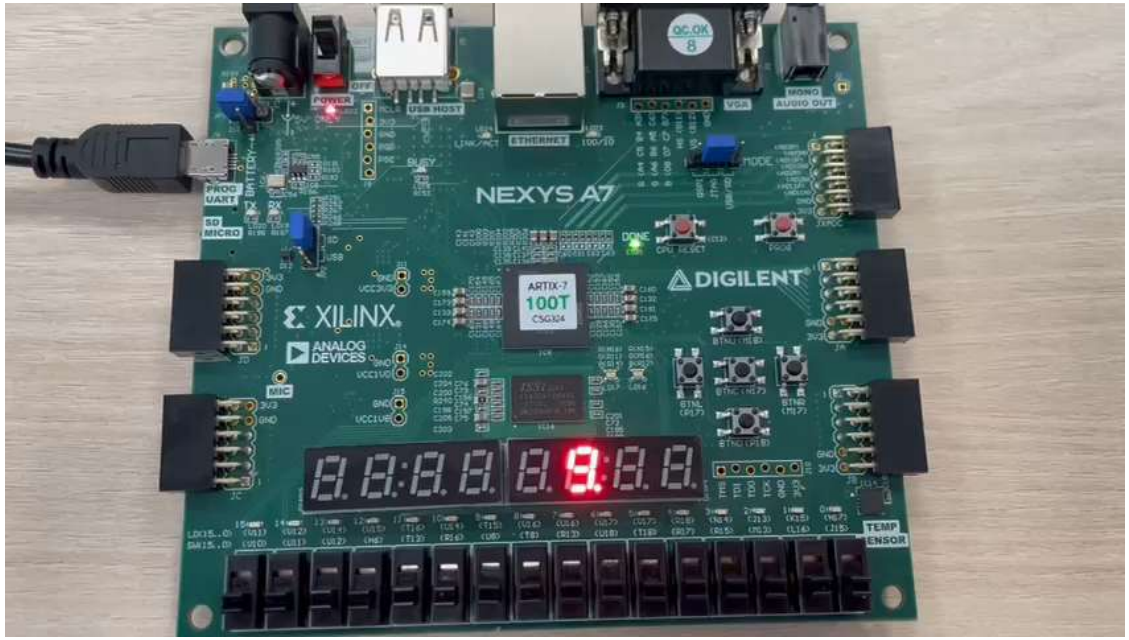


Video showing the value

1. The digits light up one by one: Frequency below 25 Hz
    - The display refresh rate is below 25/8 = 3.1 Hz

```
54      always @ * begin
55      case(RFR_DIV_SZS)
56          4'h0: EXT_DIV_RATE <= 2 * CLK_FREQUENCY_HZ; //0.5Hz
57          4'h1: EXT_DIV_RATE <= CLK_FREQUENCY_HZ; //1Hz
58          4'h2: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/2; //2Hz
59          4'h3: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/5; //5Hz
60          4'h4: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10; //10Hz
61          4'h5: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/25; //25Hz
62          4'h6: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50; //50Hz
63          4'h7: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100; //100Hz
64          4'h8: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/200; //200Hz
65          4'h9: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/400; //400Hz
66          4'hA: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1000; //1KHz
67          4'hB: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000; //10KHz
68          4'hC: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100_000; //100KHz
69          4'hD: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1_000_000; //1MHz
70          4'hE: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000_000; //10MHz
71          4'hF: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50_000_000; //50MHz
72      endcase
73      end
74
```

SW[15:12] = 1’b 1001



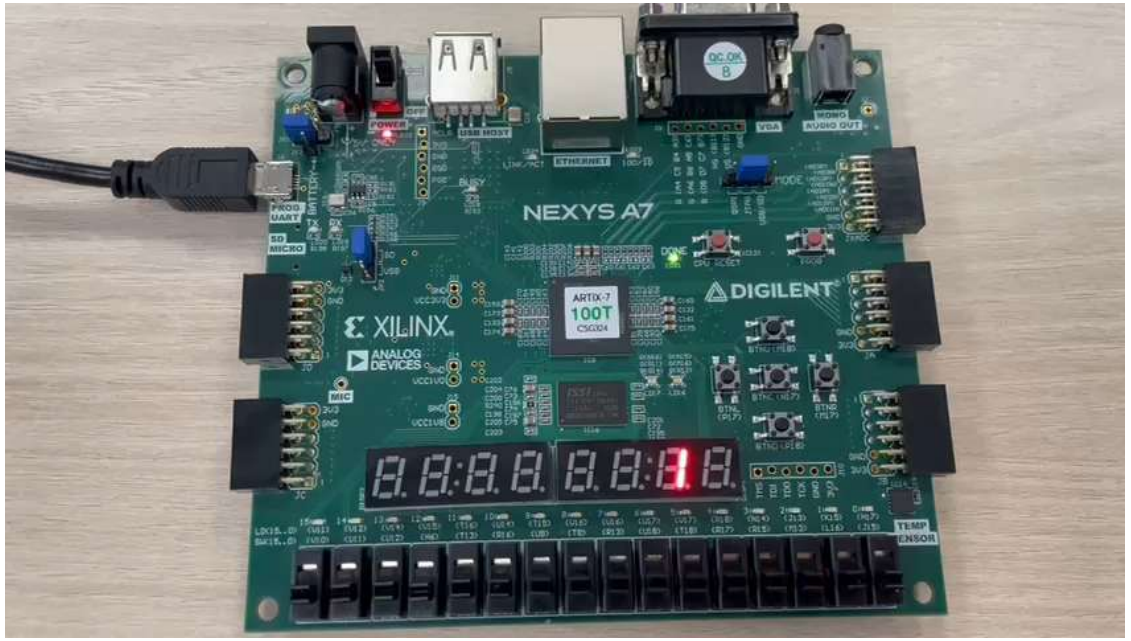
2. All the figures are visible, but with a ‘flickering’ effect:

Frequency between 50 Hz and 200 Hz

— that is, a refresh rate between 6.25 Hz and 25 HzI chose 100 Hz – that is, a refresh rate between 12.5 Hz

```
54      always @ * begin
55      case(RFR_DIV_SZS)
56          4'h0: EXT_DIV_RATE <= 2 * CLK_FREQUENCY_HZ; //0.5Hz
57          4'h1: EXT_DIV_RATE <= CLK_FREQUENCY_HZ; //1Hz
58          4'h2: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/2; //2Hz
59          4'h3: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/5; //5Hz
60          4'h4: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10; //10Hz
61          4'h5: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/25; //25Hz
62          4'h6: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50; //50Hz
63          4'h7: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100; //100Hz
64          4'h8: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/200; //200Hz
65          4'h9: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/400; //400Hz
66          4'hA: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1000; //1KHz
67          4'hB: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000; //10KHz
68          4'hC: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100_000; //100KHz
69          4'hD: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1_000_000; //1MHz
70          4'hE: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000_000; //10MHz
71          4'hF: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50_000_000; //50MHz
72      endcase
73      end
74
```

SW[15:12] = 1’b 0011

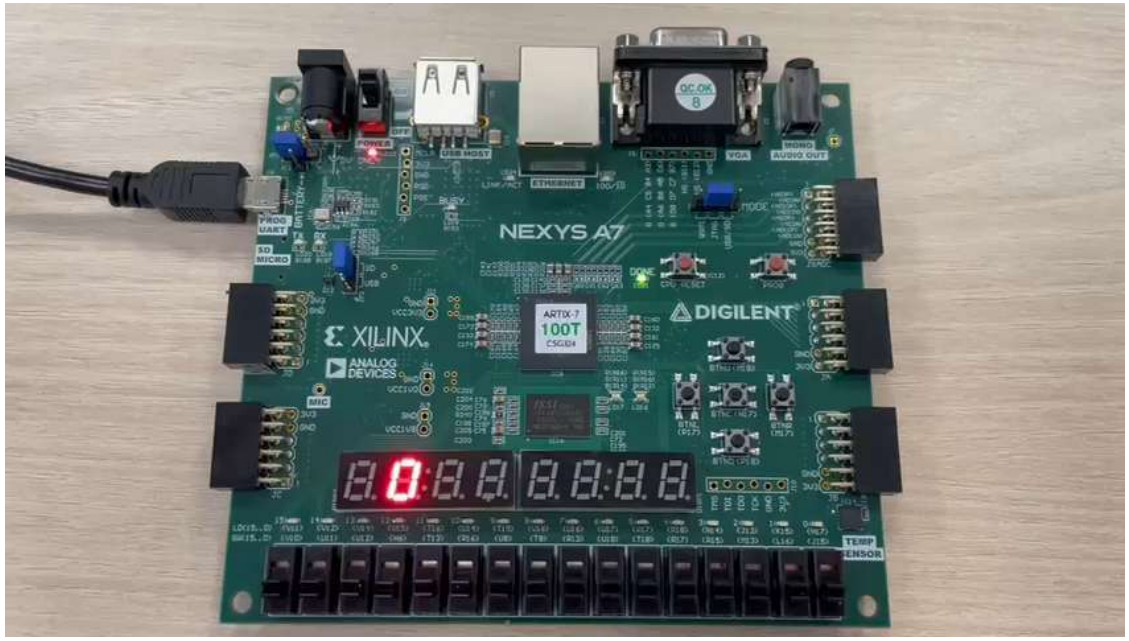


3. All the figures are visible; the display is static:

Frequency between 400 Hz and 100 kHz – that is, a refresh rate between 50 Hz and 12.5 kHz

I have selected: 10 kHz — that is, the refresh rate is 1.25 kHz

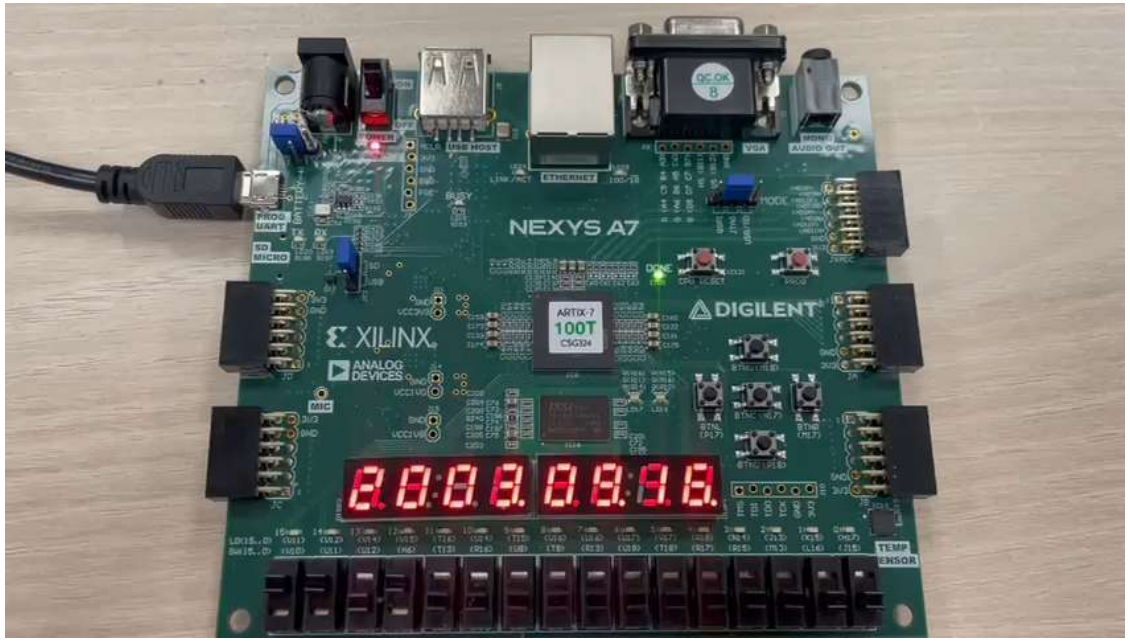
```
54      always @ * begin
55      case(RFR_DIV_SZS)
56          4'h0: EXT_DIV_RATE <= 2 * CLK_FREQUENCY_HZ; //0.5Hz
57          4'h1: EXT_DIV_RATE <= CLK_FREQUENCY_HZ; //1Hz
58          4'h2: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/2; //2Hz
59          4'h3: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/5; //5Hz
60          4'h4: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10; //10Hz
61          4'h5: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/25; //25Hz
62          4'h6: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50; //50Hz
63          4'h7: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100; //100Hz
64          4'h8: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/200; //200Hz
65          4'h9: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/400; //400Hz
66          4'hA: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1000; //1KHz
67          4'hB: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000; //10KHz
68          4'hC: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100_000; //100KHz
69          4'hD: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1_000_000; //1MHz
70          4'hE: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000_000; //10MHz
71          4'hF: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50_000_000; //50MHz
72      endcase
73      end
74
```



SW[15:12] = 1’b 1011

4-5. The figures overlap: a frequency of 1 MHz, i.e. a refresh rate of 125 kHz

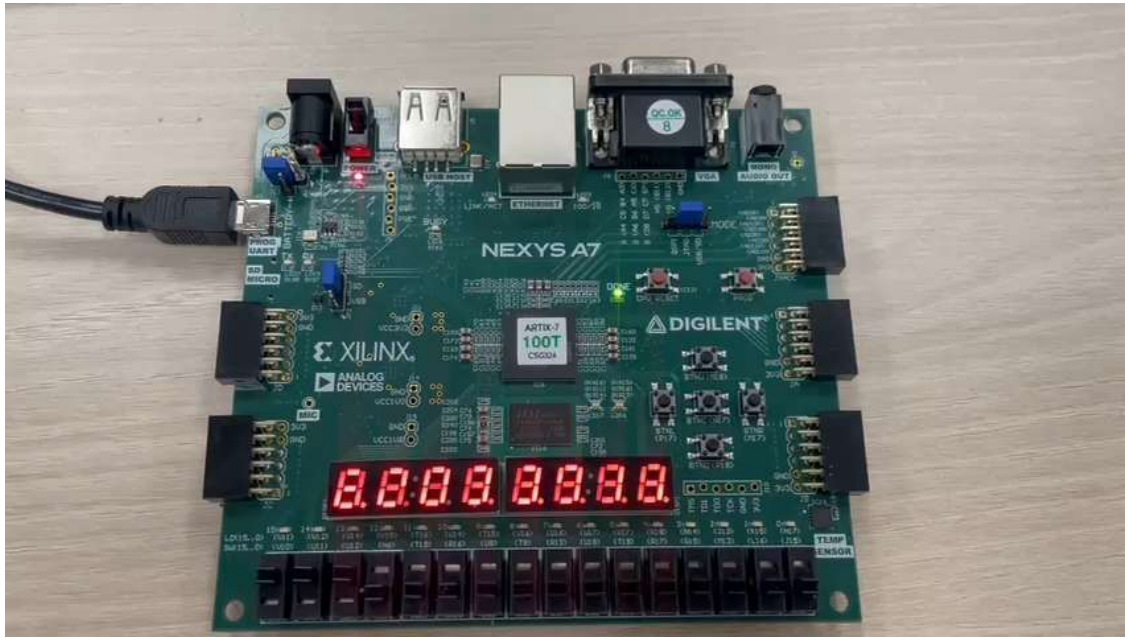
```
54      always @ * begin
55      case(RFR_DIV_SZS)
56          4'h0: EXT_DIV_RATE <= 2 * CLK_FREQUENCY_HZ; //0.5Hz
57          4'h1: EXT_DIV_RATE <= CLK_FREQUENCY_HZ; //1Hz
58          4'h2: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/2; //2Hz
59          4'h3: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/5; //5Hz
60          4'h4: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10; //10Hz
61          4'h5: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/25; //25Hz
62          4'h6: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50; //50Hz
63          4'h7: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100; //100Hz
64          4'h8: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/200; //200Hz
65          4'h9: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/400; //400Hz
66          4'hA: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1000; //1KHz
67          4'hB: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000; //10KHz
68          4'hC: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100_000; //100KHz
69          4'hD: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1_000_000; //1MHz
70          4'hE: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000_000; //10MHz
71          4'hF: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50_000_000; //50MHz
72      endcase
73      end
74
```



SW[15:12] = 1'b 1101

4-5. The figures overlap: a frequency of 10 MHz, i.e. a refresh rate of 1.25 MHz

```
54      always @ * begin
55      case(RFR_DIV_SZS)
56          4'h0: EXT_DIV_RATE <= 2 * CLK_FREQUENCY_HZ; //0.5Hz
57          4'h1: EXT_DIV_RATE <= CLK_FREQUENCY_HZ; //1Hz
58          4'h2: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/2; //2Hz
59          4'h3: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/5; //5Hz
60          4'h4: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10; //10Hz
61          4'h5: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/25; //25Hz
62          4'h6: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50; //50Hz
63          4'h7: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100; //100Hz
64          4'h8: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/200; //200Hz
65          4'h9: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/400; //400Hz
66          4'hA: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1000; //1KHz
67          4'hB: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000; //10KHz
68          4'hC: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/100_000; //100KHz
69          4'hD: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/1_000_000; //1MHz
70          4'hE: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/10_000_000; //10MHz
71          4'hF: EXT_DIV_RATE <= CLK_FREQUENCY_HZ/50_000_000; //50MHz
72      endcase
73      end
74
```

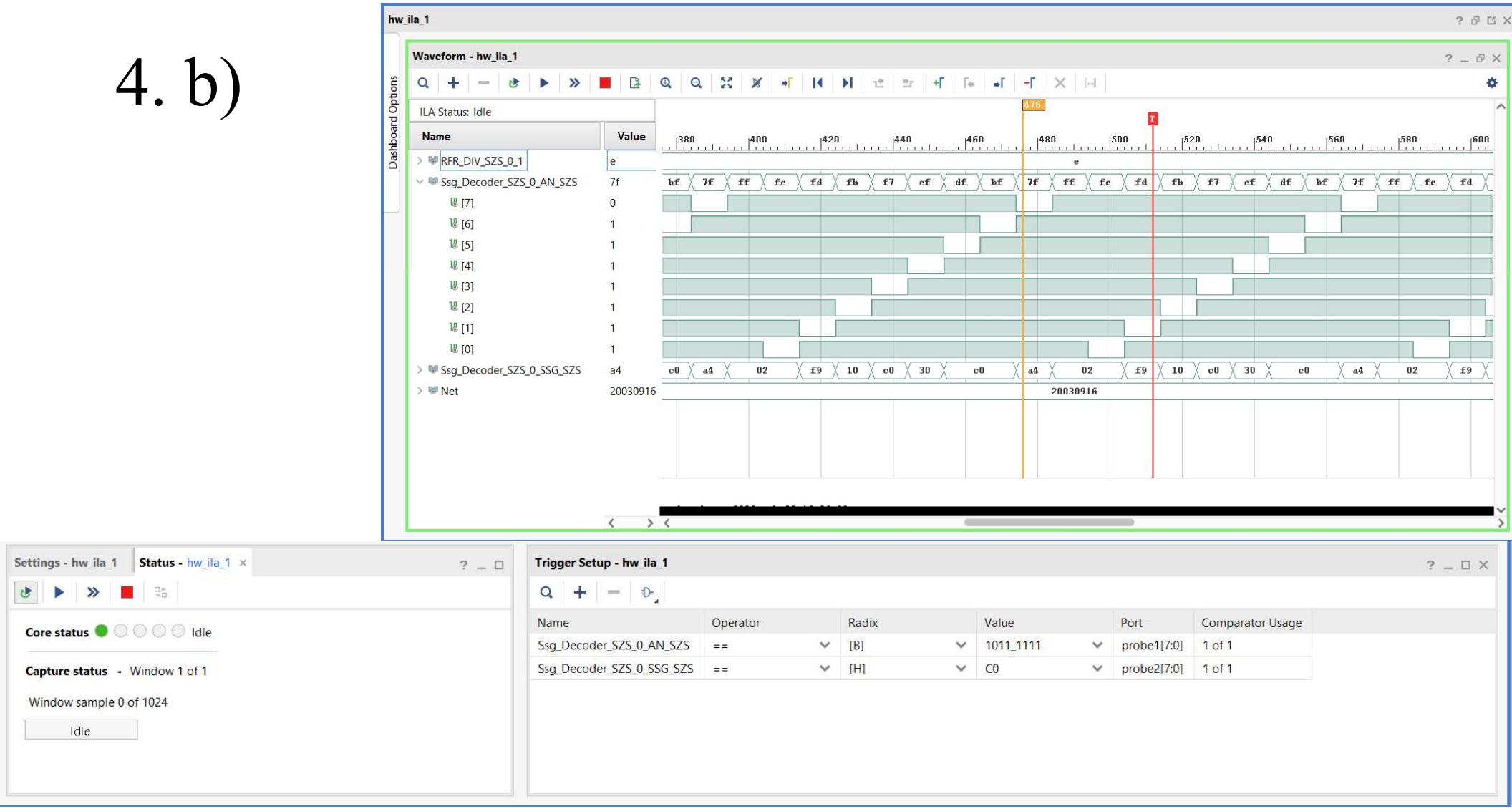


SW[15:12] = 1’b 1110

4. b) Set your ILA triggers to the following values:

- The SSG_SZS value is 8'hC0 and a Fall event occurs on the anode corresponding to the second digit from the left
- Capture an entire WALK0 period, together with the corresponding SSG codes
  - The capture must contain the trigger position (the moment of triggering)!
  - The trigger conditions must be visible on the capture!
- In the documentation, indicate for each SSG code: the corresponding digit and the decimal point value (on/reached)

4. b)


hw_ila_1
Waveform - hw_ila_1
Dashboard Options
ILA Status: Idle
Name
Value
RFR_DIV_SZS_0_1
Ssg_Decoder_SZS_0_AN_SZS
[7]
[6]
[5]
[4]
[3]
[2]
[1]
[0]
Ssg_Decoder_SZS_0_SSG_SZS
Net
20030916
Settings - hw_ila_1
Status - hw_ila_1
Core status
Idle
Capture status - Window 1 of 1
Window sample 0 of 1024
Idle
Trigger Setup - hw_ila_1
Name
Operator
Radix
Value
Port
Comparator Usage
Ssg_Decoder_SZS_0_AN_SZS
==
[B]
1011_1111
probe1[7:0]
1 of 1
Ssg_Decoder_SZS_0_SSG_SZS
==
[H]
C0
probe2[7:0]
1 of 1

6.a. The Autoblank logic code sequence

```
37
38  wire [7:0] AutoBlank_SZS, BlankDecoder;
39
40  assign BlankDecoder = (AUTOBLANK_EN_SZS)?  BLANK_SZS | AutoBlank_SZS : BLANK_SZS;
41
42  //completati autoblank ai celorlalti digiti
43
44  assign AutoBlank_SZS = (DIN_SZS[31:4] == 0)? 8'hFE  :
45                         (DIN_SZS[31:8] == 0)? 8'hFC  :
46                         (DIN_SZS[31:12] == 0)? 8'hF8 :
47                         (DIN_SZS[31:16] == 0)? 8'hF0 :
48                         (DIN_SZS[31:20] == 0)? 8'hE0 :
49                         (DIN_SZS[31:24] == 0)? 8'hC0 :
50                         (DIN_SZS[31:28] == 0)? 8'h80 : 8'h00;
51
```

6.b. Simulation results demonstrating the logic behind Autoblank

